6 Months Certification Course

Syllabus 2026

FULL STACK Development

Prepared By:
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This Full-Stack program is designed to provide complete practical knowledge of programming tools and strategies. Students will learn how to design & develop front-end and back-end systems and build successful online careers.

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WHY YOU SHOULD LEARN FULL-STACK DEVELOPMENT

Here is why you should learn full-stack development:

- **High Demand and Lucrative Career:** Full-stack developers are in high demand across industries, with companies preferring multi-skilled professionals for their ability to manage entire projects.
- **End-to-End Product Ownership:** You can handle everything from User Interface (UI) design and frontend functionality to backend logic, databases, and server management.
- **Increased Flexibility and Independence:** You can work on various parts of a project, reducing reliance on other team members and making you more adaptable in remote, fast-paced, or startup environments.
- **Greater Creativity:** You can bring your own app ideas to life from scratch, from the initial concept to the final, deployed product.

Common technologies to learn include frontend languages like HTML, CSS, and JavaScript, and backend technologies like Java, Python, Ruby, or Node.js, along with databases.

FULL STACK COURSE CONTENT

Module 1 – HTML

Introduction To Web And Basic HTML Tags

- Pieces that make the web work
- Introduction to front-end
- Basic structure of an HTML page
- Attributes, elements and relationships
- Comments
- Basic tags – Paragraph, headings, hr, br
- Basic attributes – ID and class
- Anchor tags
- Including CSS
- Including JS

Structuring Content Using HTML Tags

- Classification of elements – Block and inline
- Container tags – Div and span, and when to use them
- Semantic tags
- Including media – Images, audio and video
- Responsive media
- Lists – Ordered and unordered list
- Unstyled lists and list-based menus
- Presenting tabular data
- Styling tables

Working with Forms

- Introduction to forms
- What happens when a form is submitted
- Pieces of a simple form – Form, text input with name attribute, submit button
- Submitting data to another page (action attribute)
- Responsive media
- Changing the HTTP method used for submission (method attribute)
- Adding a label
- Default value for inputs
- Types of buttons
- Various input types – Password, textarea, checkbox, radio buttons, select dropdown, file input
- HTML5 input types – Email, URL, number, date
- HTML5 input and form attributes – Required, placeholder, novalidate

Deeper Understanding of Client-Server Communication

- Symbols in HTML
- HTTP Basics
- Introduction to chrome dev tools – The network tab
- Understanding parts of a URL
- Further exploration

Module 2 – CSS

Introduction to CSS, CSS Inheritance and Various Selectors

- Including CSS
- Choosing the way to include CSS
- User-agent styles
- Parts of a CSS rule
- Selectors – Universal, ID, class, type, attribute
- Pseudo classes – Link-related classes
- Structural Pseudo classes – :first-letter, :first-line, :first-child, :last-child, :nth-child
- Combinators – No spaces, child selector, descendant selector, selectors using comma
- Inheritance – Initial and inherit values

The Cascade, Specificity and Style Resolution

- Cascade – User-agent vs author styles
- Reset stylesheet
- Normalize Stylesheet
- Cascade of embedded styles, external stylesheets, inline styles
- Introduction to specificity – how selectors affects styles applied
- Specificity – ID vs class vs type selector
- Specificity calculation and style resolution in case of complex selectors
- Use of !important
- Units for CSS properties – United and unitless, absolute vs relative
- Default values for properties (initial values)
- Computed and resolved styles

Important CSS Properties

- Box model in depth
- Various ways of specifying colors
- Background properties
- Using background properties with image sprites
- Typography related properties
- Handling overflow
- Hiding elements

Deeper Understanding of Client-Server Communication

- Introduction to responsive web design (RWD)
- Fluid layouts
- Fluid images
- Media queries
- Further exploration

Module 3 – JavaScript

Introduction, Variables, Scopes & Data Types

- History of JavaScript
- Setting up the environment
- Running JavaScript in the browser and Node
- Comments
- Variables and primitive data types
- Falsy and truthy values
- Variable scopes, scope chain
- Using arrays
- The type of operator

Operators, Control Flow & Functions

- Variable hoisting
- Operators and expressions
- Control flow – Branching and looping
- Function declaration and usage
- Anonymous function and function expression
- IIFE
- Function hoisting
- Functions call context (this keyword)
- Handling variable number of arguments
- Callbacks – passing functions as arguments

Functions (Continued) & Objects

- Returning functions
- Object declaration using literal syntax
- Accessing properties and methods
- Adding and deleting properties
- Listing object properties – for..in loop and object.keys()
- Constructor function and the new keyword
- Function – functions as objects
- call(), apply() and bind() as methods of functions

Built-in Objects and Functions

- Array methods
- Date methods
- JSON
- Number methods
- String methods

Exception Handling & Browser Objects

- Strict mode execution
- Error objects
- Exception handling
- Window
- Navigator
- Location
- History

DOM and Event Handling

- The document object
- Nodes and the DOM tree
- Node relationships and DOM tree traversal
- Methods for DOM manipulation
- Various browser events
- Different ways to handle events
- Event object properties and methods

Introduction to ES6

- Introduction, setting up, running ES6 code, JS transpilers
- Scope, let & const
- Template literals, default parameters
- Destructuring arrays & objects
- Rest & spread

More ES6

- Arrow functions
- Classes & inheritance
- Modules
- Promises

Module 4 – TypeScript

Basics of TypeScript

- Introduction to TypeScript
- Why TypeScript?
- Setting up TypeScript
- JavaScript vs TypeScript
- Type annotations, variable declarations, basic datatypes, type inference
- Advanced types, type erasure and error behavior
- Classes, constructors & methods, inheritance & polymorphism, access modifiers

Deeper Dive into TypeScript

- Interfaces, properties & methods, interfaces & classes, extending interfaces
- Namespaces, namespaces using multiple files
- Modules, import, export, namespaces vs modules
- Generics, generic functions, generic classes

Introduction to Angular

- What is Angular?
- Why Angular?
- Angular versions
- Where does Angular fit?
- Multi page application (MPA)
- Single page application (SPA)
- Setting up Angular
- Create your first Angular app
- Serve your Angular app
- Edit your first Angular component

Project Structure, Modules, Bootstrapping

- Understand project structure
- Modules
- Decorators
- Bootstrapping
- Add Bootstrap to our app
- Use Bootstrap in our app

Data Binding and Component Interaction

- Data binding
- Component interaction
- Angular forms: Template driven forms & reactive forms

Directives, Pipes, Services and Dependency Injection

- Directives
- Pipes
- Services
- Dependency injection

Routing and Communicating with Server

- Building SPAs using routing
- Understanding observables
- Server communication using Http Client
- Error handling
- Implementing authentication in Angular

Module 5 – Programming

Coding Fundamental and Basics of Programming

- Basics of programming
- Programming environment
- Fundamental keywords
- Basic operators
- Decision making operators
- Loop statements
- Functions in the program
- File I/O

Concept of OOPs

- OOPS
- Abstraction
- Encapsulation
- Inheritance
- Polymorphism
- Classes
- Interfaces
- Enum

Control Statements

- If statement
- While statement
- For and Do-while
- Continue
- Break

I/O and Arrays

- Sending input to program
- Displaying output
- Displaying output as error
- Arrays
- One dimensional arrays
- Two dimensional arrays

String, Math, Formatters etc.

- Strings
- Identifying length of string
- String comparing
- Searching within string
- Replacing string
- Math functions
- Formatting strings

Packages

- Naming conventions in packages
- Creating package
- Importing all classes in package

Exception Handling

- Why exception handling
- Keywords in exception handling
- Identifiers
- Try-catch
- Throw
- Finally

Collections

- Goals of collection
- Collection interfaces
- List
- Iterator
- Map

Files & Threads

- Reading a file
- Writing a file
- Appending to existing file
- Check file exists
- Delete file, asynchronous process
- What is thread
- Run a method parallelly

Streams

- Streams
- forEach
- IntStream
- Map
- Filter
- Limit
- Skip
- Collect

Date & Time API

- Date & tim e objects
- Manipulate date operations
- Date formatters

Module 6 – Backends Databases

- MongoDB vs RDBMS
- Install MongoDB & Compass
- Databases
- Collections
- CRUD document
- Projection
- Sort, skip & limit
- Indexing & aggregation
- Backup & restore

DevOps & Cloud deployment

- Jenkins
- Git
- Packaging & distributing
- Deployment on AWS

Certificate

Upon completing this Certification course, you will receive a program completion certificate from Tech Engineer, Mohali (Pb). This certificate will testify to your skills as an expert in the completed course.

Contact Us

Visit Our HQ

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